

Exploratory Data Analysis (EDA) Report on Titanic Dataset

Objective

The objective of this project was to perform Exploratory Data Analysis (EDA) on the Titanic dataset using Python. The analysis aimed to identify patterns, trends, and relationships among passenger attributes such as age, gender, passenger class, fare, and survival status.

Tools Used

Python, Pandas, Matplotlib, Seaborn, and Jupyter Notebook.

Dataset Overview

The Titanic dataset contains passenger information including Passenger Class (Pclass), Sex, Age, Fare, and Survived status. The dataset is commonly used for learning data analysis and machine learning concepts.

Steps Performed

1. Loaded the dataset using Pandas.
2. Examined dataset structure using `info()` and `describe()`.
3. Checked missing values using `isnull()`.
4. Used `value_counts()` to analyze categorical variables.
5. Created histograms, boxplots, scatterplots, pairplots, and heatmaps.
6. Identified trends and relationships affecting passenger survival.

Findings and Observations

1. Survival Analysis

The analysis showed that the number of passengers who did not survive was greater than the number of survivors. This indicates that survival was limited during the disaster.

2. Gender-Based Survival

Female passengers had a significantly higher survival rate than male passengers. This suggests that gender played an important role in rescue priority.

3. Passenger Class Analysis

Passengers traveling in First Class had better survival rates compared to those in Second and Third Class. Third-Class passengers experienced the highest mortality rate.

4. Age Distribution

The histogram of Age showed that most passengers were between 20 and 40 years old. There were fewer elderly passengers and children in the dataset.

5. Fare Analysis

The boxplot of Fare revealed several outliers, indicating that some passengers paid significantly higher ticket prices. Passengers with higher fares generally showed better survival chances.

6. Correlation Analysis

The correlation heatmap highlighted relationships between numerical variables. Passenger Class and Fare showed an important relationship with survival outcomes.

Conclusion

The Exploratory Data Analysis of the Titanic dataset revealed that gender, passenger class, and fare were the most important factors influencing survival. Female passengers and First-Class passengers had the highest survival rates. The project provided practical experience in data exploration, visualization, and extracting meaningful insights from data.